Consolidated Technical Compliance & Safety Report: Vehicle Firmware Suite

Parameter	Specification
Project	DYNAMICS UPC - Formula Student Electric Vehicle Firmware Suite
Audited Firmware Targets	**MCU** (Motor Control Unit), **ECU** (Electronic Control Unit), **PDM** (Power Distribution Module)
Production Codebase	`MCU/MCU_FW/`, `ECU/ECU_FW/`, `PDM/PDM_FW/` (ESP-IDF v5 + FreeRTOS SMP)
Applied Standards	MISRA-C:2012, ISO 26262 (ASIL-D / ASIL-B Principles), Zero Dynamic Memory, 100 Hz Deterministic FreeRTOS
CI/CD Build & Test Status	**PASSED** (Compilation with `-Werror`, static analysis, and 100% host unit tests passing)

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1. Executive Summary & Distributed System Architecture

This report certifies the functional safety, industrial compliance, and quality metrics of the embedded firmware suite powering the **DYNAMICS UPC** Formula Student electric racing vehicle.

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graph TD
    subgraph Sensors & Driver Inputs
        APPS[Dual Redundant APPS] --> MCU
        HPS[Hydraulic Brake HPS] --> MCU
        ENC[4x IRAM Wheel Encoders] --> MCU
        STEER[Steering Angle Pot] --> MCU
        NTC[Bosch NTC Thermistors SPI] --> ECU
        STS[4x Suspension Strain Gauges] --> ECU
        SHUNTS[12x Shunts + 2x Hall Sensors] --> PDM
    end

    subgraph Deterministic CAN Bus Network (500 kbps)
        MCU[MCU / VCU Core] <-->|0x020, 0x021, 0x502| BUS[Differential CAN Bus]
        ECU[ECU Cooling Control] <-->|0x401, 0x402, 0x503| BUS
        PDM[PDM LV Power Hub] <-->|0x001..0x006, 0x501| BUS
        BUS <-->|0x0C0 Torque Cmd| INV[Unitek Bamocar Inverter]
        BUS <-->|Telemetry Logging| DASH[Dashboard & Data Logger]
    end

    subgraph Actuators & Power Distribution
        ECU -->|LEDC PWM 14-bit 50 Hz| ESC[Motor & Inverter ESCs]
        PDM -->|12x Solid-State Gate Drivers| MOSFETS[LV Loads: Pumps, Sensors, ECU, Steering Wheel]
        MCU -->|Torque Command & R2D Enable| MOTOR[Electric Traction Motor]
    end
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2. Consolidated Vehicle Fault Management Matrix (DTC & Safe States)

Every anomaly across the vehicle is categorized, assigned deterministic hardware safe-state actions, and broadcasted as standard Diagnostic Trouble Codes (DTC):

Board	DTC Code	Fault Identifier	Cat.	Prio.	Trigger Condition	System & Hardware Reaction	Recovery / Reset	Broadcasted CAN Frame
ECU	`201`	`FAULT_CODE_MOTOR_NTC_FAIL`	`HW`	`HIGH`	≥ 3 failed reads on Motor NTC.	Ramps Motor cooling fan (+10%/s) to **100% continuous power**.	Automatic when both NTCs read valid temperatures.	ID `0x503` (DTC `201`, Fans at 100%)
ECU	`202`	`FAULT_CODE_INV_NTC_FAIL`	`HW`	`HIGH`	≥ 3 failed reads on Inverter NTC.	Ramps Inverter cooling fan to **100% continuous power**.	Automatic once valid temperature returns.	ID `0x503` (DTC `202`)
ECU	`203`	`FAULT_CODE_ADS8688_SPI_ERR`	`HW`	`HIGH`	SPI communication timeout with ADS8688 ADC.	Invalidates telemetry; forces both fans to **100%**.	Requires SPI bus reset or hard reboot.	ID `0x503` (DTC `203`)
ECU	`204`	`FAULT_CODE_TWAI_BUS_OFF`	`COMM`	`HIGH`	CAN controller enters `BUS_OFF` state.	Calls `twai_initiate_recovery()` and restarts CAN driver.	Automatic upon physical bus recovery.	ID `0x503` upon recovery
PDM	`10..21`	`FAULT_CODE_OVERCURRENT_CH(0..11)`	`HW`	`HIGH`	Current on shunted channel i > 1.30 × I_{nom} (fast trip or 3 samples for CH3/CH9).	**Turns off MOSFET pin (`HIGH`)**, forces status to 0, latches channel in `fault_manager`, and rejects CAN overrides.	Requires vehicle power cycle or authorized reset.	IDs `0x001` / `0x002` (Status=0), ID `0x501` (DTC 10+i, Locked Mask)

Board	DTC Code	Fault Identifier	Cat.	Prio.	Trigger Condition	System & Hardware Reaction	Recovery / Reset	Broadcasted CAN Frame
PDM	`100`	`FAULT_CODE_VBAT_UNDERVOLTAGE`	`HW`	`HIGH`	Battery voltage V_{bat} < 5.0V for > 200 ms.	**Cuts power to all 12 MOSFET channels** to protect battery cells against destructive deep discharge.	Automatic when V_{bat} > 5.0V on power-up.	IDs `0x001` / `0x002` (All 0), ID `0x006` (V_{bat}), ID `0x501` (DTC `100`)
PDM	`1`	`FAULT_CODE_CAN_ERROR_PASSIVE`	`COMM`	`LOW`	Alert `ERR_PASS` due to bus degradation.	Internal diagnostic logging and telemetry warning.	Automatic when error rate decreases.	ID `0x501` (DTC `1`)
PDM	`2`	`FAULT_CODE_CAN_BUS_OFF`	`COMM`	`HIGH`	CAN bus error count > 50 or `BUS_OFF` .	Initiates automatic hardware CAN recovery cycle.	Automatic upon bus resynchronization.	ID `0x501` upon recovery
MCU	`101`	`FAULT_CODE_APPS_IMPLAUSIBLE`	`HW`	`HIGH`	Discrepancy between APPS1 and APPS2 > 10\% for > 100 ms.	**Immediately clamps torque to 0.0 Nm**, latches APPS subsystem, and disables inverter command.	Automatic when sensor discrepancy drops below 10\%.	ID `0x502` (DTC `101` , APPS Mask Locked)
MCU	`102`	`FAULT_CODE_APPS_WIRE_BREAK`	`HW`	`HIGH`	Analog APPS voltage outside ± 15\% calibrated bounds.	**Immediately clamps torque to 0.0 Nm** and blocks drive.	Restoration of valid analog voltage range.	ID `0x502` (DTC `102`)
MCU	`103`	`FAULT_CODE_BRAKE_SENSOR_ERR`	`HW`	`HIGH`	Hydraulic pressure sensor disconnected or out of electrical bounds (< 50 or > 4000).	Latches brake subsystem, blocks R2D entry, limits torque.	Restoration of valid electrical reading.	ID `0x502` (DTC `103` , Brakes Mask Locked)
MCU	`104`	`FAULT_CODE_TWAI_BUS_OFF`	`COMM`	`HIGH`	Inverter CAN or Car CAN in `BUS_OFF` state.	Initiates recovery cycle and forces 0.0 Nm torque.	Automatic upon physical bus recovery.	ID `0x502` upon recovery
MCU	`105`	`FAULT_CODE_BMS_SAG_LIMIT`	`RES`	`LOW`	HV battery pack voltage drops near minimum threshold during acceleration.	Dynamic torque derating via internal resistance (R_{int}) and OCV estimator to prevent BMS shutdown.	Dynamic based on open-circuit voltage.	ID `0x502` (DTC `105`)
MCU	`106`	`FAULT_CODE_BSPD_TRIPPED`	`HW`	`HIGH`	Accelerator > 25\% with brake engaged (> 100 ADC).	**Hard clamp of torque demand to 0.0 Nm** (FS EV4.7).	**Latched**: Clears only when accelerator pedal is released to < 5\%.	ID `0x502` (DTC `106`)

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3. Global Vehicle CAN Bus Communication Matrix (500 kbps)

CAN ID	Message Name	Sender	Receivers	DLC	Rate	Payload Description
`0x001`	`PDM_MOSFETS_1_8`	**PDM**	Dashboard, MCU, Telemetry	8	10 Hz	Status of MOSFET channels 1 through 8 (1 = ON, 0 = OFF).
`0x002`	`PDM_MOSFETS_9_12`	**PDM**	Dashboard, MCU, Telemetry	4	10 Hz	Status of MOSFET channels 9 through 12 (1 = ON, 0 = OFF).
`0x003`	`PDM_CURRENTS_0_3`	**PDM**	Telemetry, Data Logger, MCU	8	10 Hz	Current measurements for channels 0..3 (LE uint16 mA).
`0x004`	`PDM_CURRENTS_4_7`	**PDM**	Telemetry, Data Logger, MCU	8	10 Hz	Current measurements for channels 4..7 (LE uint16 mA).
`0x005`	`PDM_CURRENTS_8_11`	**PDM**	Telemetry, Data Logger, MCU	8	10 Hz	Current measurements for channels 8..11 (LE uint16 mA).
`0x006`	`PDM_CURRENTS_HALL_VBAT`	**PDM**	Telemetry, Data Logger, MCU	8	10 Hz	Hall SD (mA), Hall Fans (mA), LV Battery Voltage (mV), Steering Wheel Alert.
`0x020`	`MCU_WHEEL_SPEEDS`	**MCU**	Dashboard, Telemetry, Logger	8	100 Hz	Wheel speeds for FL, FR, RL, RR (BE uint16 RPM).

CAN ID	Message Name	Sender	Receivers	DLC	Rate	Payload Description
`0x021`	`MCU_VEHICLE_STATE`	**MCU**	Dashboard, ECU, PDM, Telemetry	8	100 Hz	Steering angle, brake pressures, R2D state (4 = R2D), and demanded torque.
`0x0C0`	`INVERTER_TORQUE_CMD`	**MCU**	Unitek Bamocar Inverter	8	100 Hz	Torque command register `0x90` and demanded value (0 \dots 32767).
`0x100`	`MANUAL_MOSFET_CMD`	**Steering Wheel**	PDM	2	On-Event	Channel ID (0 \dots 11) and power toggle command (1 = ON, 0 = OFF).
`0x200`	`MCU_LOG_TELEMETRY`	**MCU**	Data Logger, Telemetry	8	10 Hz	Diagnostic logging and controller telemetry.
`0x401`	`ECU_TEMPS`	**ECU**	MCU, Dashboard, Telemetry	4	1 Hz	Motor and Inverter temperatures (BE int16, 1^°C/LSB).
`0x402`	`ECU_STS_GAUGES`	**ECU**	Telemetry, Dynamics Logger	8	100 Hz	Raw 16-bit counts from 4 suspension travel strain gauges (RR, RL, FR, FL).
`0x501`	`PDM_DIAGNOSTIC_DTC`	**PDM**	Safety Master, Logger, Dash	8	10 Hz / Event	High Fault Active, Category, Priority, Active DTC, Locked Channels Bitmask.
`0x502`	`MCU_DIAGNOSTIC_DTC`	**MCU**	Safety Master, Logger, Dash	8	10 Hz / Event	High Fault Active, Category, Priority, Active DTC, Locked Subsystems Bitmask.
`0x503`	`ECU_DIAGNOSTIC_DTC`	**ECU**	Safety Master, Logger, Dash	8	10 Hz / Event	Thermal Failsafe Active, Active DTC, Motor Fan %, Inverter Fan %.

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4. Unified Unit Testing & CI/CD Verification Suite

All 15 test cases across the three firmware packages execute in native x86 environments and achieve 100% test pass rates in GitHub Actions CI:

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ECU_FW Unity Test Suite:
test/test_main.cpp:88: test_ads8688_bosch_ntc_conversion      [PASSED]
test/test_main.cpp:89: test_fan_driver_esc_scaling            [PASSED]
test/test_main.cpp:90: test_fan_driver_slew_rate              [PASSED]
test/test_main.cpp:91: test_pid_cooling_reverse_action        [PASSED]
test/test_main.cpp:92: test_fault_manager_failsafe_escalation [PASSED]
----------------------- 5 Tests 0 Failures 0 Ignored (PASSED) ------------------

PDM_FW Unity Test Suite:
test/test_main.cpp:95: test_vbat_conversion                   [PASSED]
test/test_main.cpp:96: test_protection_undervoltage_debounce  [PASSED]
test/test_main.cpp:97: test_protection_overcurrent_fast_trip  [PASSED]
test/test_main.cpp:98: test_protection_inrush_debouncing      [PASSED]
test/test_main.cpp:99: test_fault_manager_channel_lock        [PASSED]
----------------------- 5 Tests 0 Failures 0 Ignored (PASSED) ------------------

MCU_FW Unity Test Suite:
test/test_main.cpp:95: test_apps_calibration_and_deadband     [PASSED]
test/test_main.cpp:96: test_bspd_interlock                    [PASSED]
test/test_main.cpp:97: test_r2d_state_machine                 [PASSED]
test/test_main.cpp:98: test_torque_modes_and_limits           [PASSED]
test/test_main.cpp:99: test_fault_manager_apps_implausibility [PASSED]
----------------------- 5 Tests 0 Failures 0 Ignored (PASSED) ------------------
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5. Industrial Quality & Safety Certification

1. **Zero Dynamic Memory Allocation**: Prohibits heap allocation (`malloc` , `free` , `new` , `delete`) across all production firmware modules. All tasks, queues, and buffers are 100% statically allocated at compile time.
2. **Hard Real-Time FreeRTOS Determinism**: 100 Hz primary execution loops utilizing `vTaskDelayUntil` for zero-jitter timing.
3. **MISRA-C:2012 Automotive Compliance**: Strict integer typing, explicit arithmetic conversions, bounded arrays, and zero uninitialized variables.
4. **Zero Warnings Enforcement**: Continuous compilation under `-Wall -Wextra -Werror` in ESP-IDF v5 and native GCC environments.